ARRCUS
NETWORK DIFFERENT™

Shepherding the 5G and Edge Computing Era with Open, Software-powered Networking Infrastructure Solutions

Tetsuya Murakami
tetsuya@arrcus.com



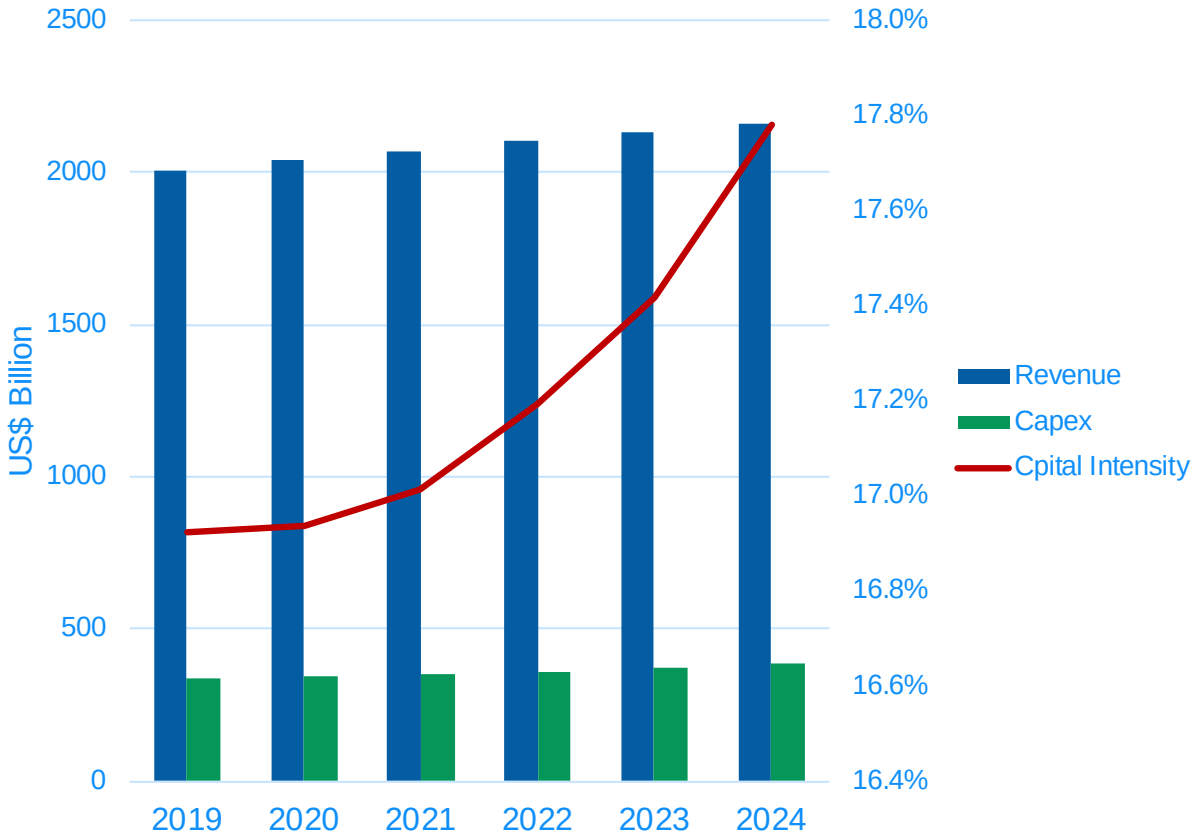
Agenda

- Service Provider Challenges Moving to 5G
- Networking Trends
- Scaling 5G Networks
- Virtualized Distributed Router

Arrcus, Inc Confidential - Light Reading Webinar

Service Providers Face Revenue Crunch

Communications Service Provider Revenue and Capital Intensity, 2019-2024



- Global CSP Revenue:
  - 1.5% CAGR 2019-2024
- Global CSP Capex:
  - 2.5% CAGR 2019-2024
- Capital Intensity:
  - 2019: 16.9%
  - 2024: 17.8%

Note: Includes fixed and mobile operators. Excludes Internet content providers and content neutral providers.
Source: Omdia Communications Provider Revenue & Capex Forecast Highlights: 2019–2024, August 2019

Arrcus, Inc Confidential - Light Reading Webinar

What Keeps Operators Up at night?

Operator Network Priorities (Mobile and Converged Only)

Network Priority	Overall Rank
Adding to top-line revenue	1
Meeting demand for bandwidth growth	2
Preparing my network for 5G	3
Protecting network infrastructure	4
Reducing opex via automation	5
Preparing my network for edge compute	6

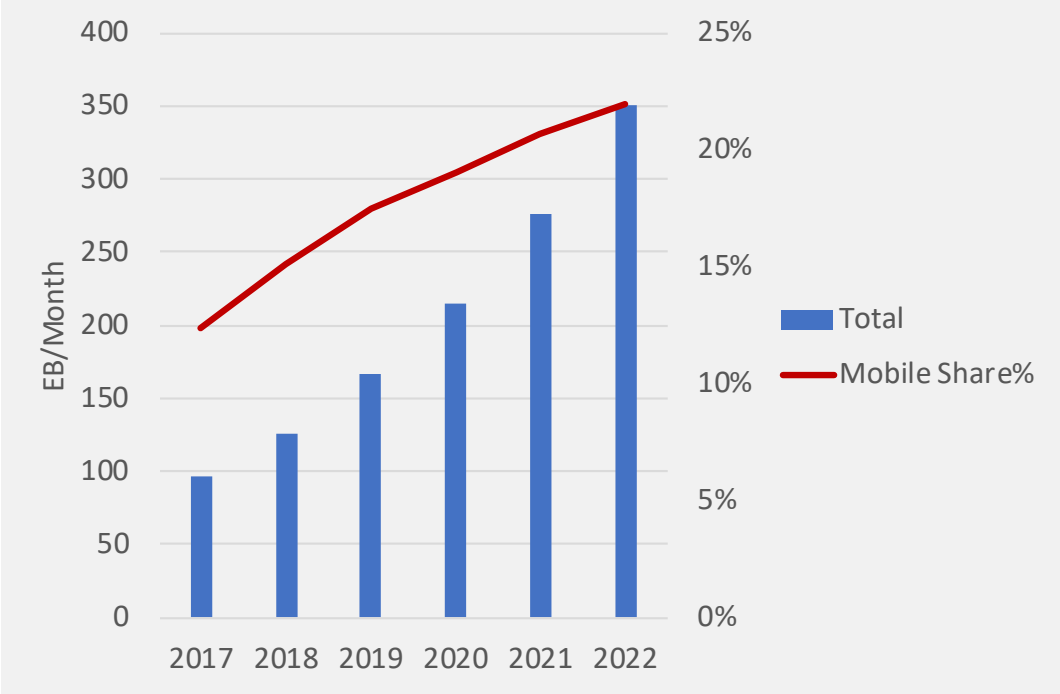
N=58

Source: Heavy Reading Future of IP Networking Survey, Sponsored by Cisco, 2019

Mobile Revolution Also Poses Challenges

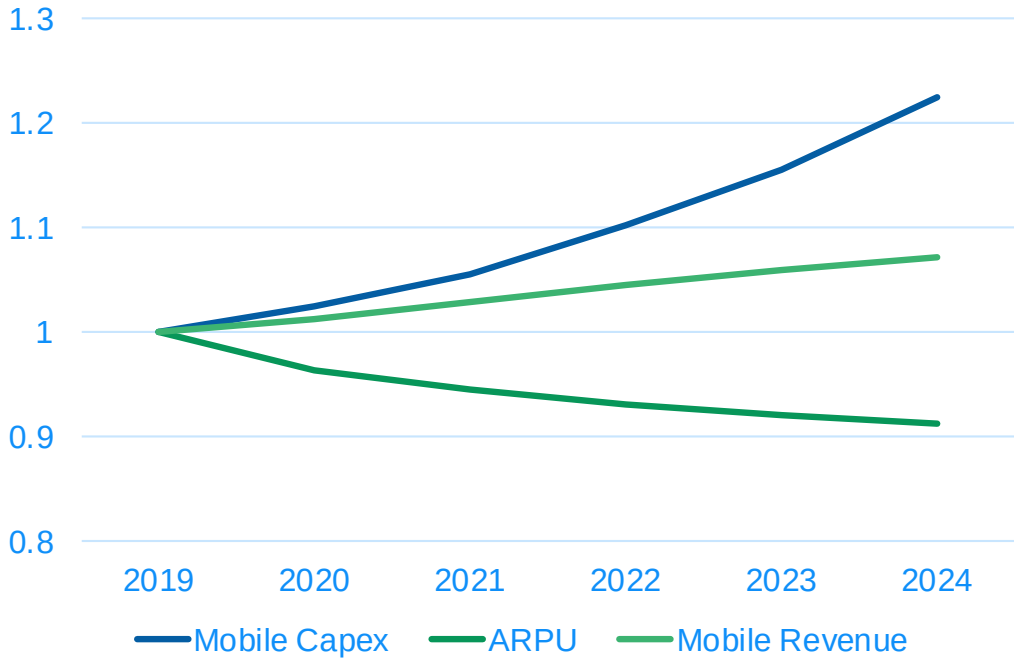
- Internet traffic: 29% CAGR
- Mobility impacts cannot be ignored

Global Internet Traffic and Mobile Share, 2017-2022



Note: Excludes Managed IP Traffic, Source: Cisco VNI, 2019

Growth in global mobile revenue, ARPU, and capex: 2019–24



Source: Omdia, March 2020

- Mobile capex rising faster than revenue
- Global ARPUs on the decline

Arrcus, Inc Confidential - Light Reading Webinar

5G Network Challenges : Transport Perspective

- Meeting ultra-low latency requirements (network + edge strategy)
- Extending wireline connectivity to new cells sites (cell site densification)
- Achieving RAN capacity requirements
- Meeting network synchronization requirements
- Maintaining 4G networks while also adding 5G networks
- And doing all the above with minimal increase to capex!

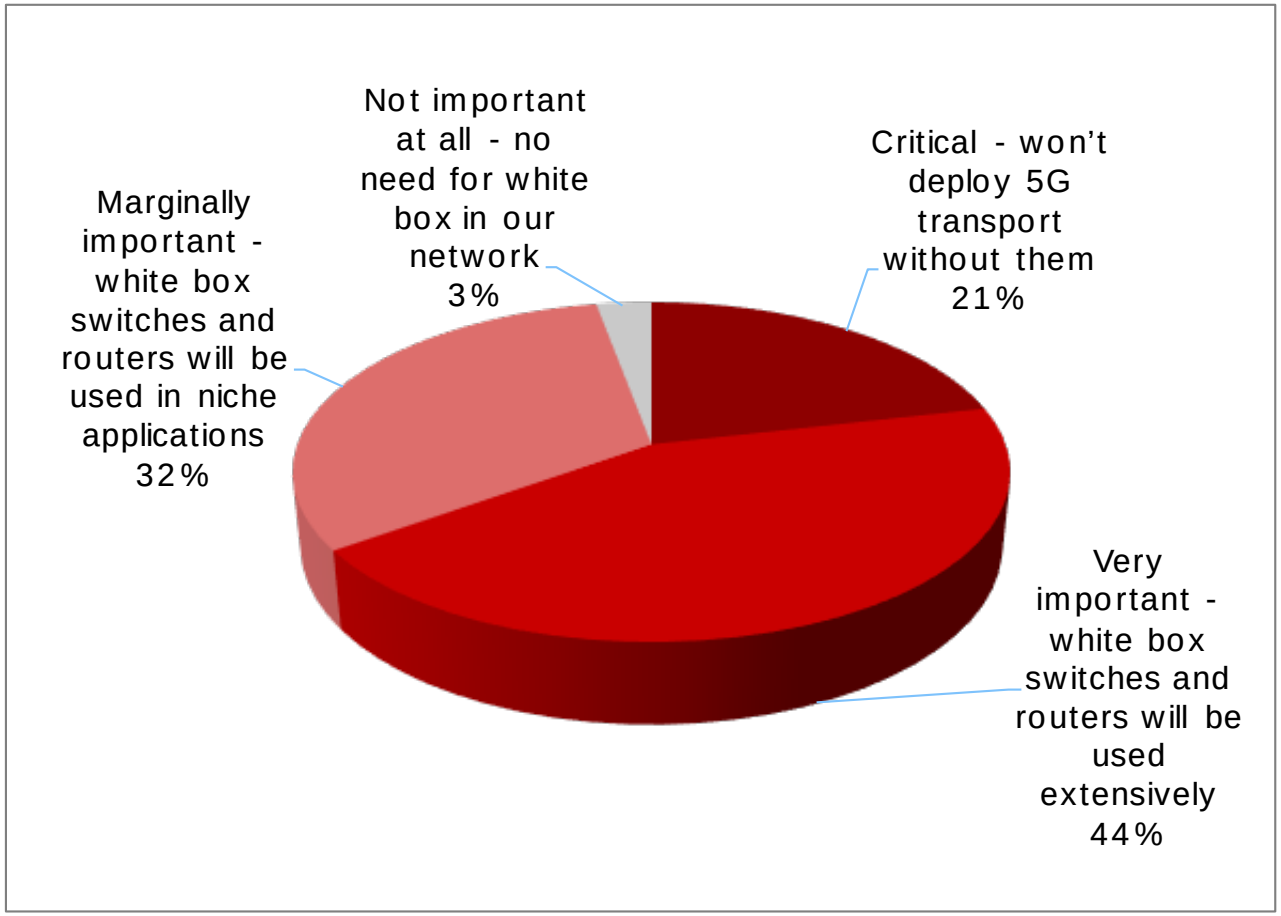
Arrcus, Inc Confidential - Light Reading Webinar

Operators Look to Disaggregation at the IP Layer

Disaggregation Definition

- The separation of networking equipment into **functional components** and allowing each component to be individually deployed:
  - Encompasses separation of software OS from underlying hardware
  - Requires open APIs to enable SDN control
- Degrees of disaggregation exist
  - Not one size fits all in the market
- **White Box** is a specific subset of disaggregation that requires open spec hardware produced by contract manufacturers
  - White box has its own benefits and challenges

Importance of White Box Switches / Routers in 5G



N=103
Source: Heavy Reading

ARRCUS
NETWORK DIFFERENT™

“Getting into the Groove”



Legacy Solutions Yet Ever-Increasing Challenges

INTERNET USERS
2023: 5.3B
+36%

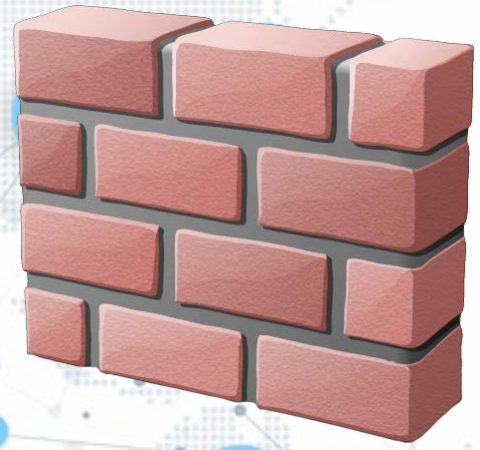
NETWORKED DEVICES
2023: 29.3B
+60%

M2M CONNECTIONS
2023: 14.7B
+141%

MOBILE DATA TRAFFIC
2023: 100EB/Mo.
+233%

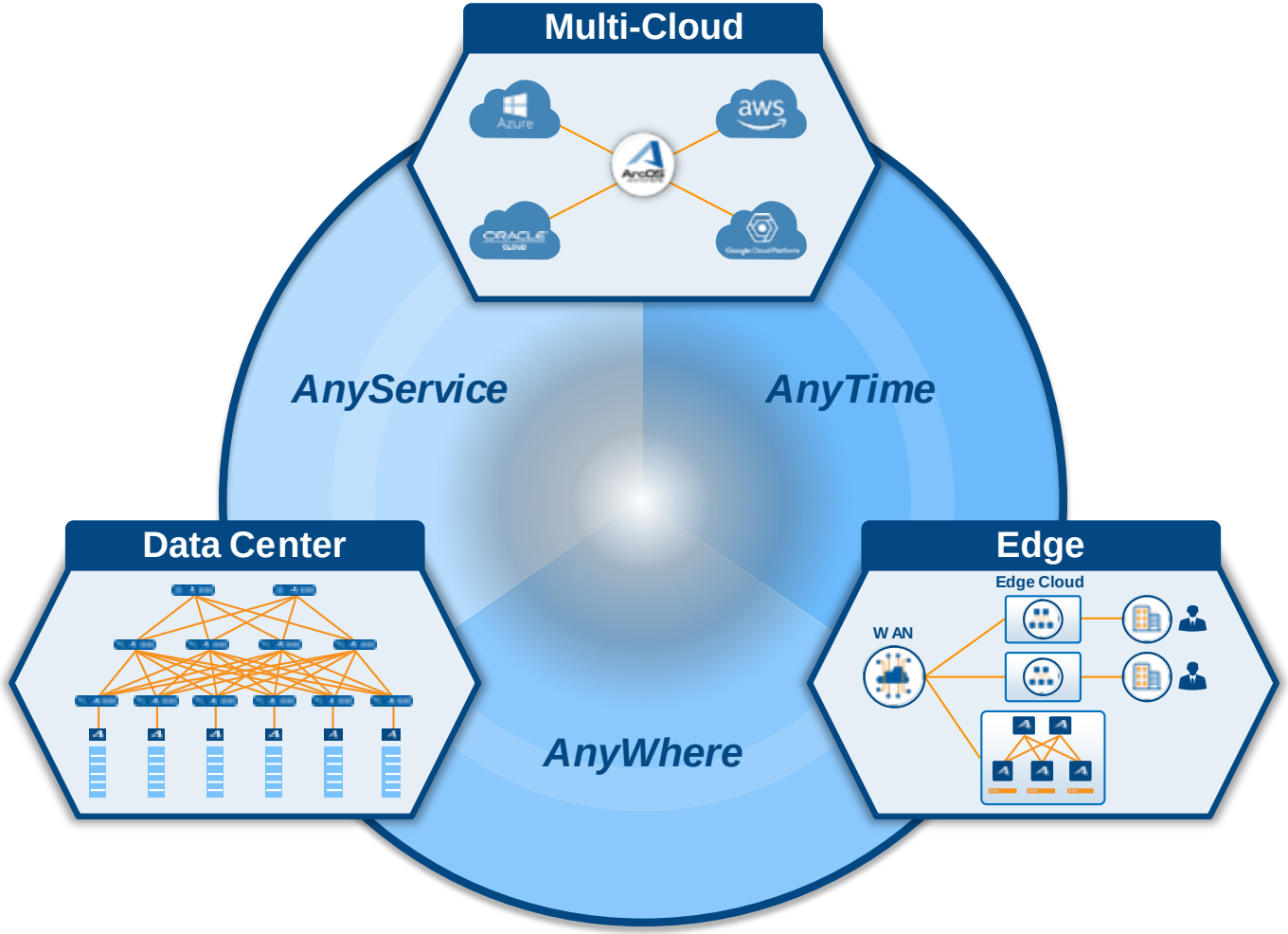
~$60B

LEGACY OEM ~ANNUAL REVENUES



WHY USE THE SAME OLD SOLUTIONS?

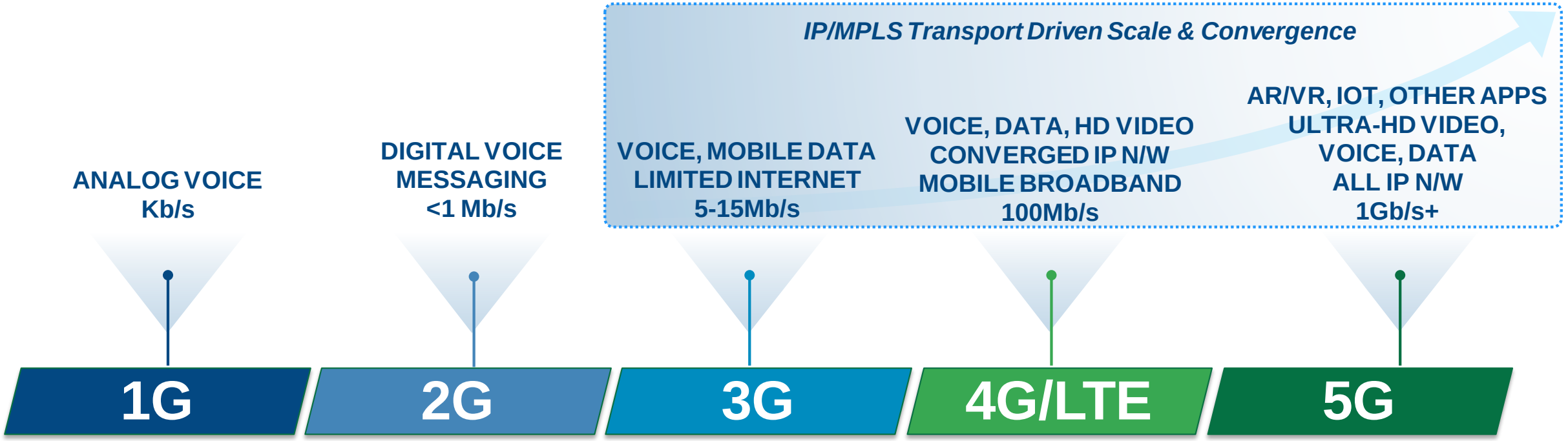
Demand of the Hyperscale Networking Software



Enterprise. CSPs. Multi-Cloud.

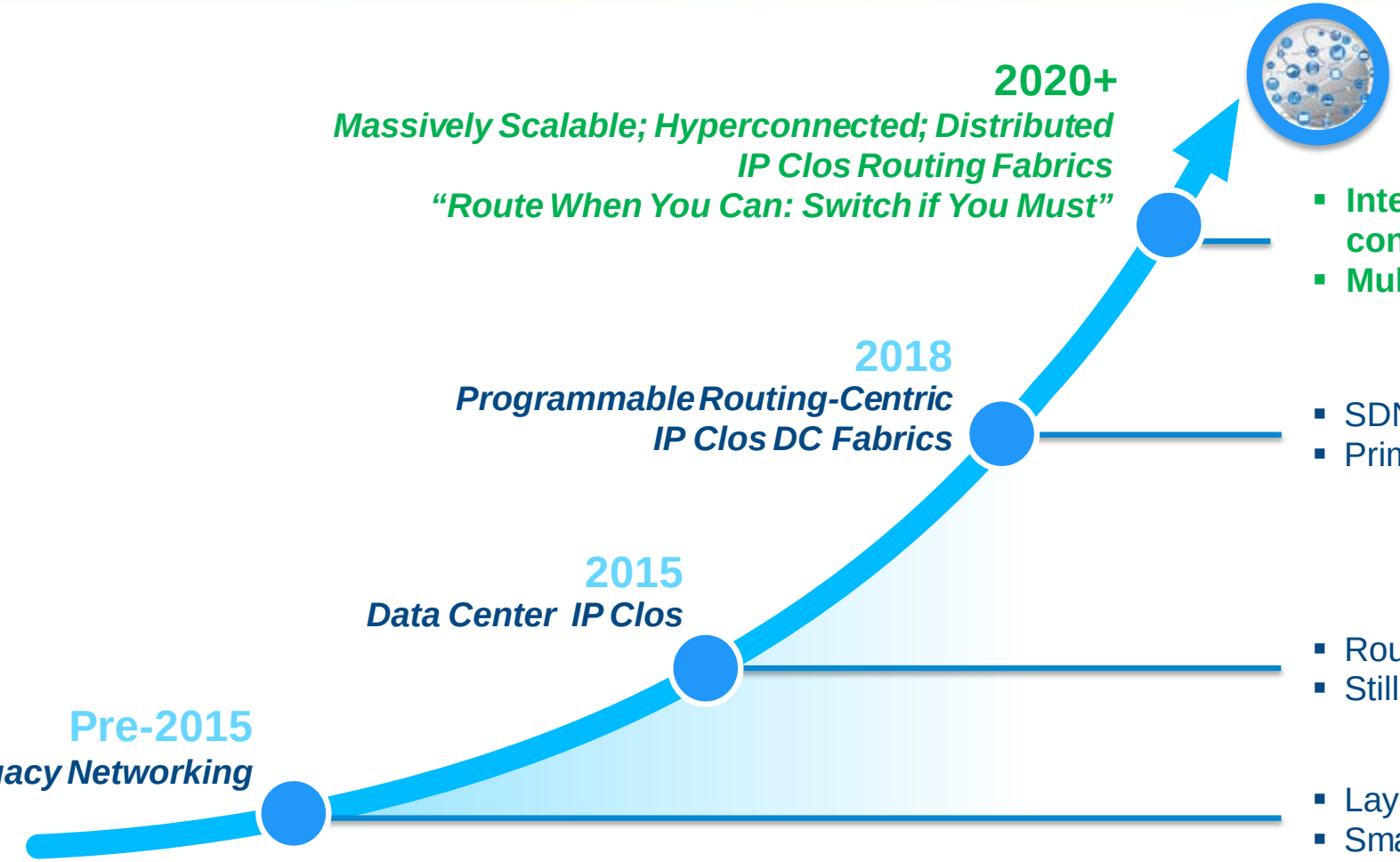
Arrcus, Inc Confidential - Light Reading Webinar

IP Networking Fuels Scale, Performance, & Resilience Of Mobile Infrastructure



Arrcus, Inc Confidential - Light Reading Webinar

5G, Multi-Cloud & Edge Requires Routing Intelligence



2020+

Massively Scalable; Hyperconnected; Distributed IP Clos Routing Fabrics

"Route When You Can: Switch if You Must"

- **Intelligent connectivity, centralized policy & control, distributed data plane**
- **Multi-Cloud, 5G Edge, EPC, CO, Store Fronts…**

2018

Programmable Routing-Centric IP Clos DC Fabrics

- SDN Multi-Tenant Routing to ToR (EVPN, SR)
- Primarily DC, switching-routing convergence

2015

Data Center IP Clos

- Routing to ToR
- Still largely limited to Cloud Titans

Pre-2015

Legacy Networking

- Layer 2-centric
- Small control plane PoDs

The Control Plane is the Brains of Your Network

Software is Eating Networking

5G, Multi-Cloud & Edge Market Trends

Digital Network Transformation

Continuous Next

Business Drivers



Enhanced Experiences



New Monetization Models



Control Costs

Networking Imperatives



SRv6, SR-MPLS, VPN Hybrid



Five 9s SLA, Resilient, Secure Low-Latency



Scale-Up; Scale-Out, Composable Software Programmable; HW Agnostic

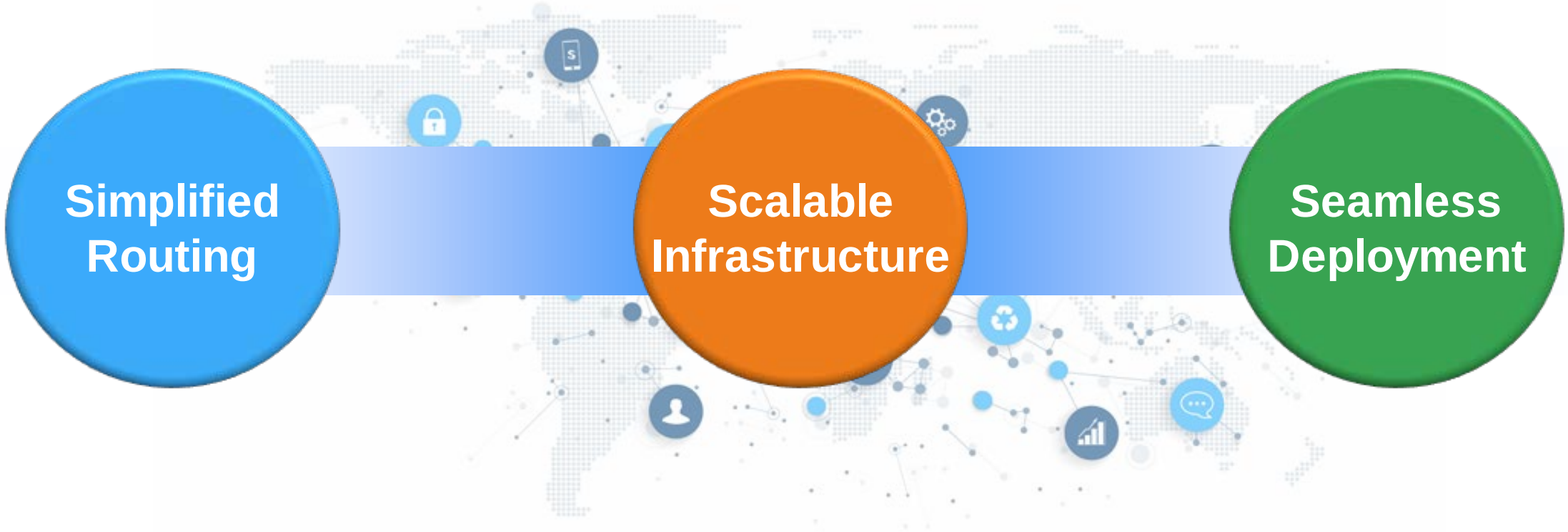
AnyService. AnyTime. AnyWhere

Networking Innovations are Driving 5G Infrastructures

5G Tenets	Networking Enablers
HYPERSCALE ECONOMICS	Lean, Scalable, & Composable Networking SW; Hardware Agnostic
TECHNOLOGICAL SCALE	1G/10G/100G/400G/800G speeds, Hyperscale Control Plane
LOW LATENCY	Virtualized Distributed Routing, CUPS (No Network Tromboning)
NETWORK SLICING	SR (MPLS,v6); End-to-End Logical VPN Topologies
FLEXIBILITY	Converged SDN Programming, Automated Path Computation
RELIABILITY	End-to-End QoS, PTPv2 / IEEE 1588

Arrcus, Inc Confidential - Light Reading Webinar

5G: Towards a Simple, Scalable, & Seamless Network



- EVPN w/L2VPN & L3VPN
- SR w/MPLS/v6 Data Planes
- Standards-based APIs

- Lean, Resilient, Composable Software
- SDN-Programmable Infra, NFV
- White Box Economics

- Across DCs, Multi-Cloud & Mobile Edge
- Automated Control-Plane Policy Functions
- Distributed User-Plane Functions

Arrcus, Inc Confidential - Light Reading Webinar

Traditional Approaches Cannot Meet the Needs of Modern Networks



- Rigid
- Brittle
- Expensive
- Limited Scale
- Opaque
- Proprietary/Lock-in

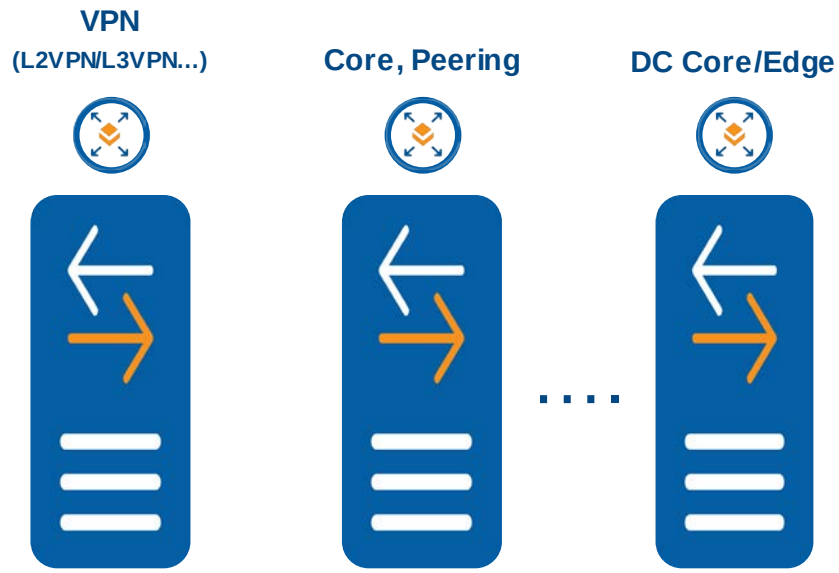
LEGACY

- Flexible
- Resilient
- Pay-as-you-grow
- Hyperscale
- Deep Visibility
- Open

MODERN

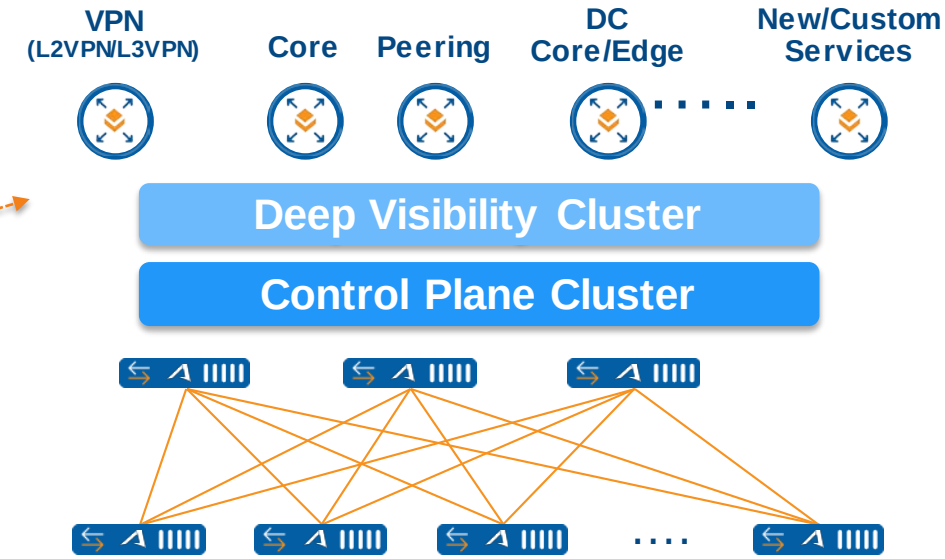
Drive Massive Flexibility & Scale of Network Services

Traditional Approach: Vertical Scaling



- Limited Scale
- Monolithic
- Opaque

Horizontal Auto-Scaling

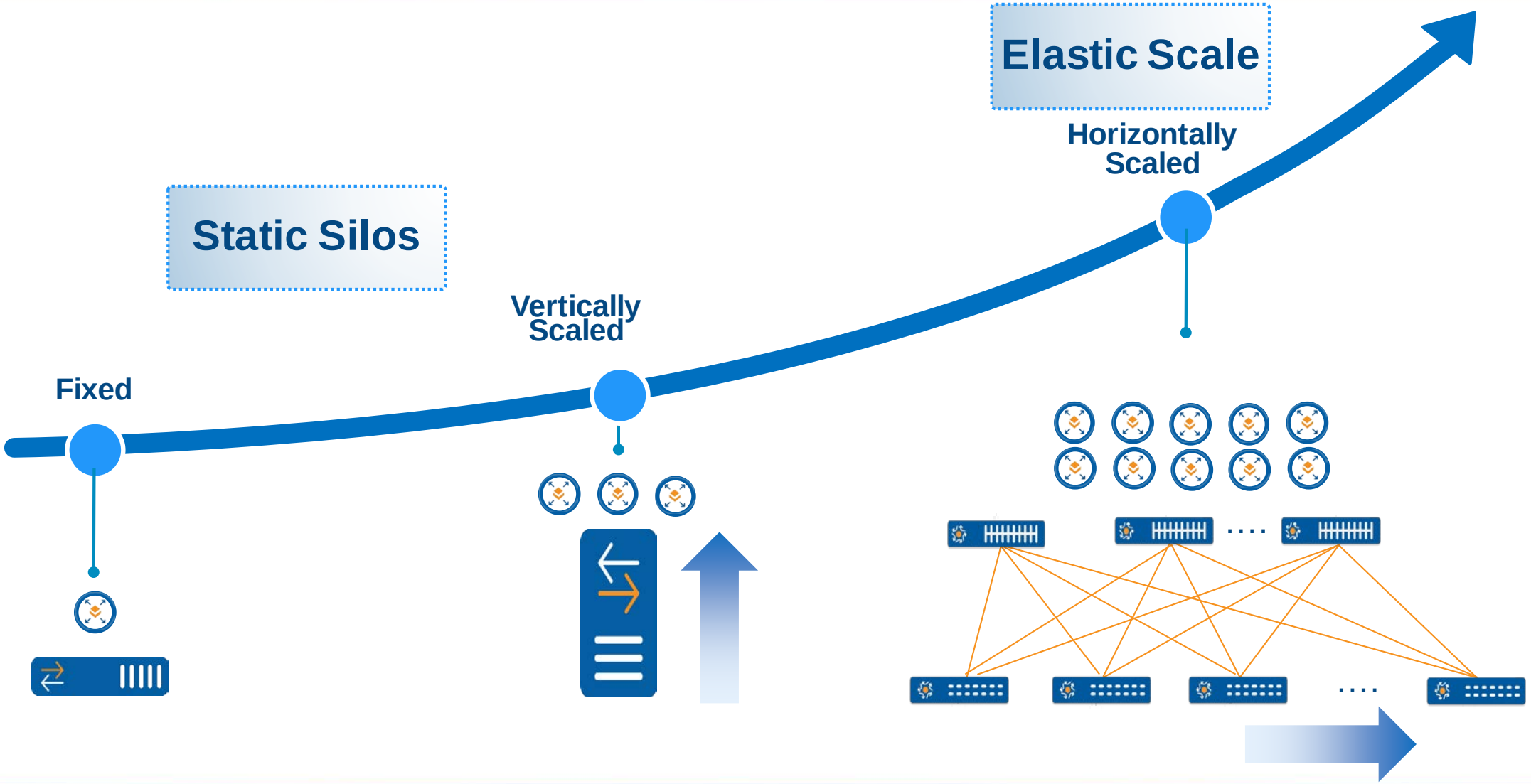


- Massive Scale
- Virtualized, Distributed
- Deep Visibility

Services & Applications

Arrcus, Inc Confidential - Light Reading Webinar

5G Innovations Drive Hyperscale Architectures to IP/MPLS Transport Networks



Arrcus, Inc Confidential - Light Reading Webinar

Segment Routing is a Key Building Block of 5G

High-Performance, Resilient Networks

- Sub 50 msec Fast Re-route
- No Extra Protocol Overhead: LDP or RSVP-TE
- Significantly Reduced State/Tunnels w/Scale(vs. GTP)

Application-Centric Multi-Service Networks

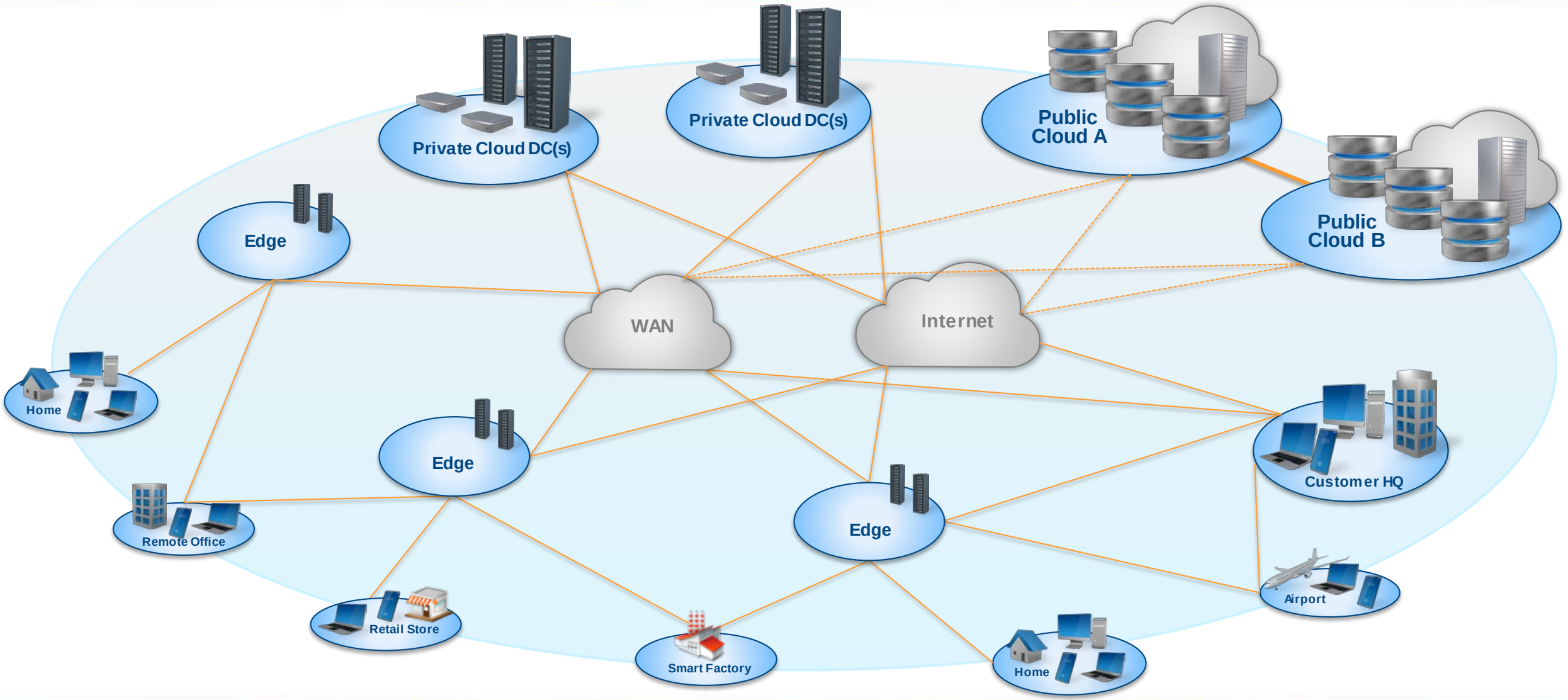
- App SLAs Dictate Slice Composition
- Enables Application Engineered Routing (AER)
- AER Facilitates 5G Use Cases/Business Models

Software-Defined, Adaptive Networks

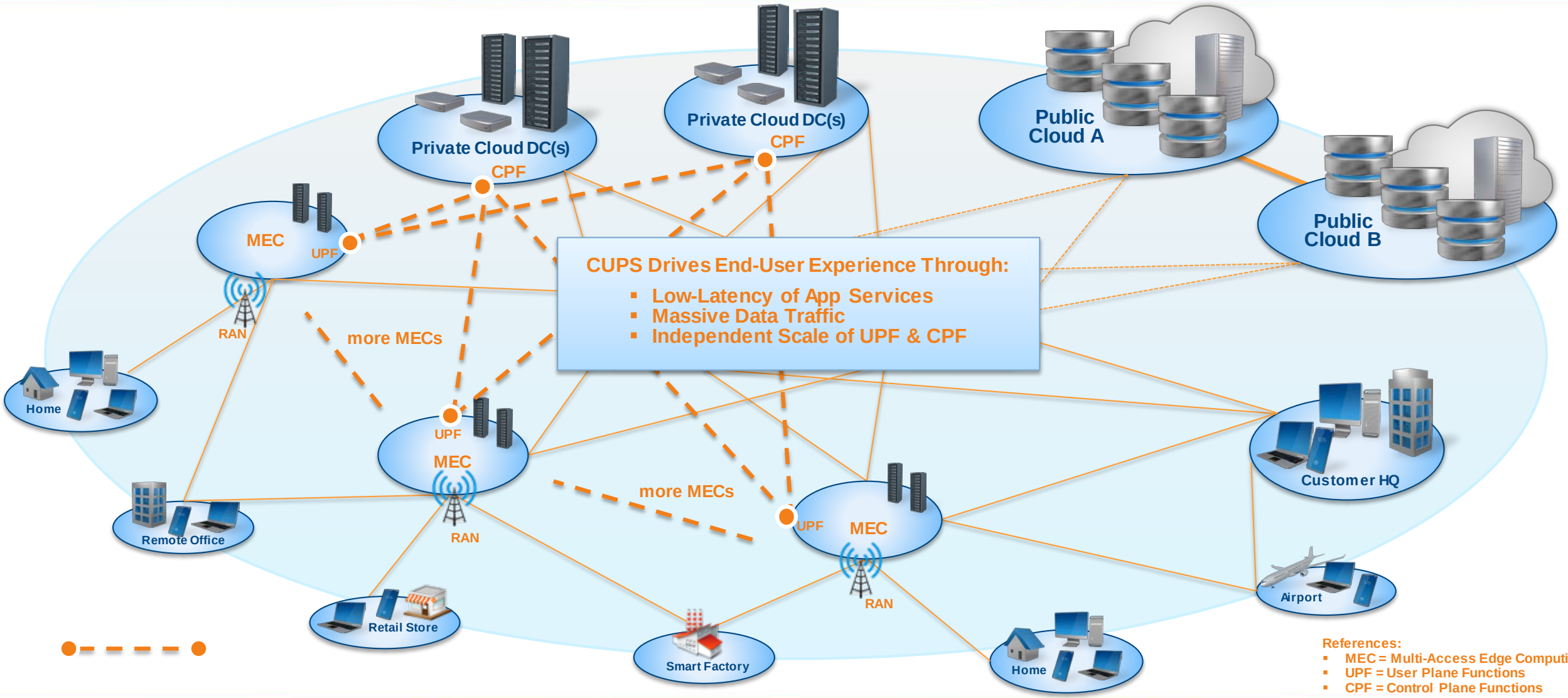
- SR-TE Drives Dynamic Traffic Re-Routing
- Maximum, Efficient Network Utilization
- Only SR Delivers TE at Massive Scale

SR-MPLS / SRv6

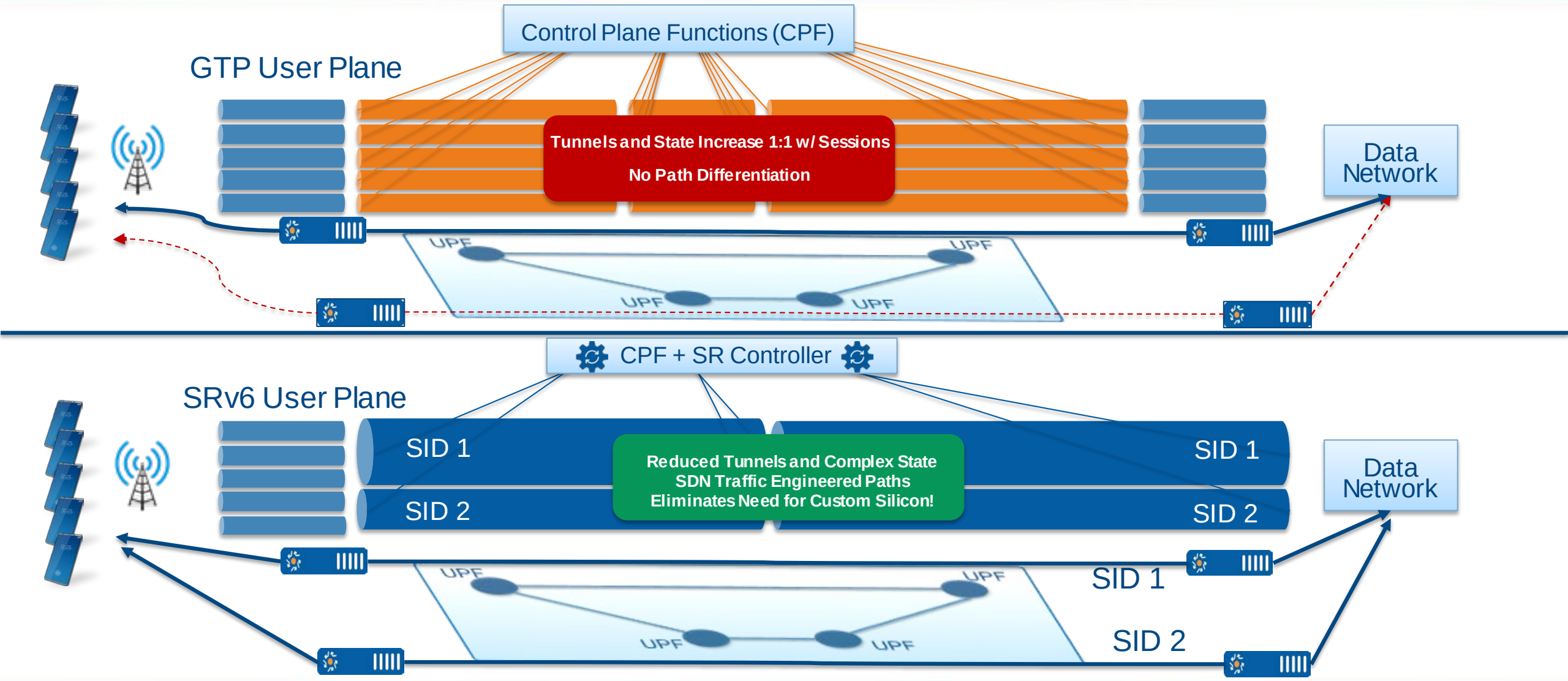
SDN-Centric, Virtualized, Distributed Routing Enables 5G Control & User Plane Separation (CUPS)



SDN-Centric, Virtualized, Distributed Routing Enables 5G Control & User Plane Separation (CUPS)

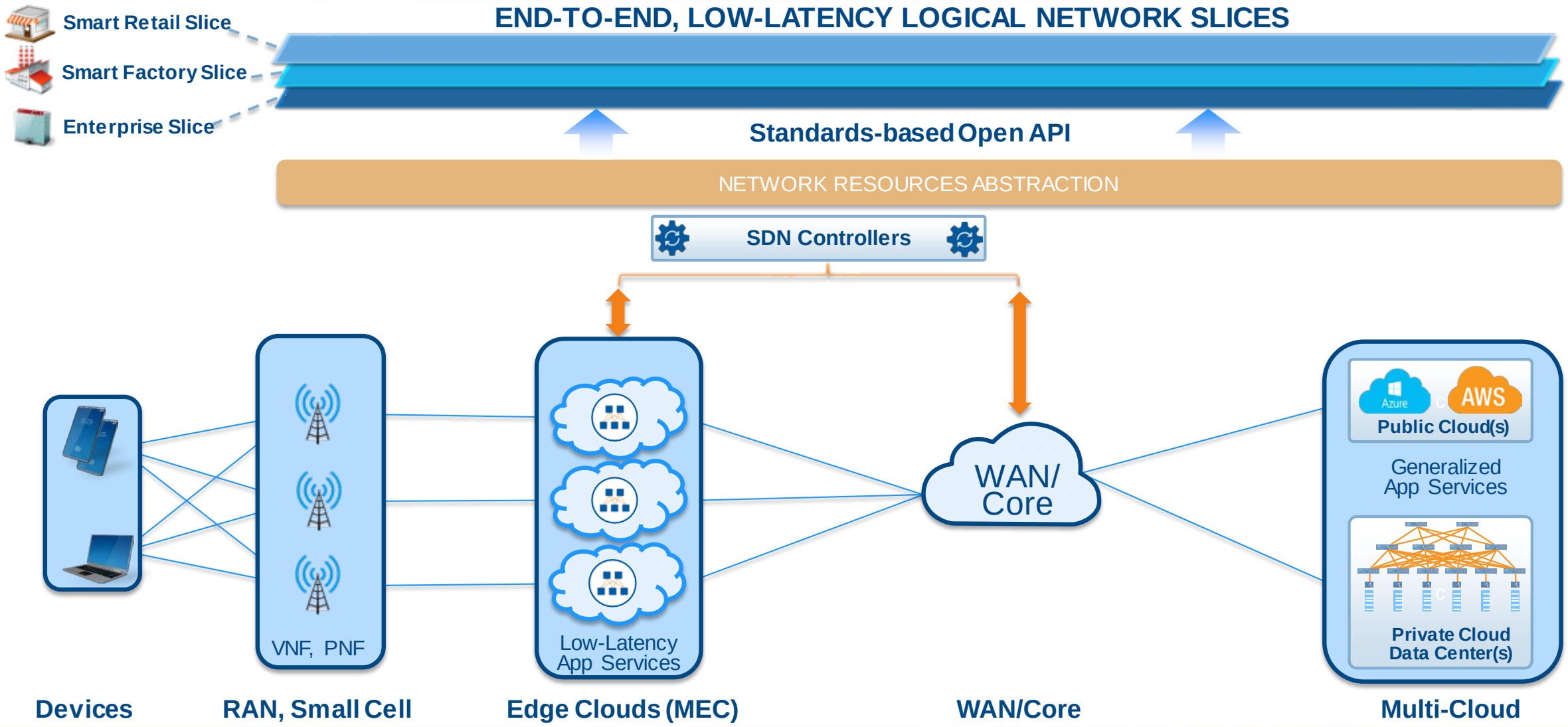


Reduced Complexity of 5G Networks using Segment Routing



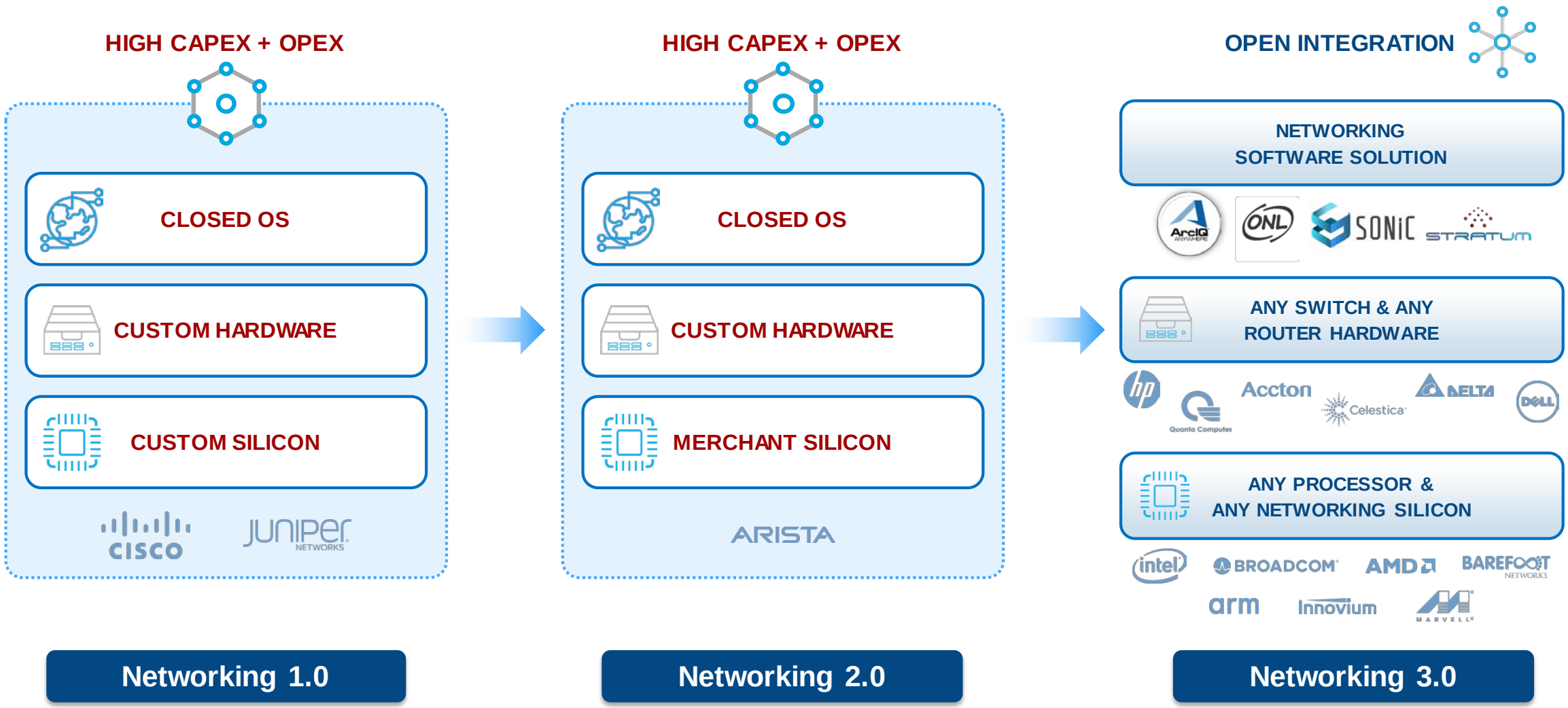
Arrcus, Inc Confidential - Light Reading Webinar

Scale 5G Services with End-to-End Multi-Tenant Logical Networks



Arrcus, Inc Confidential - Light Reading Webinar

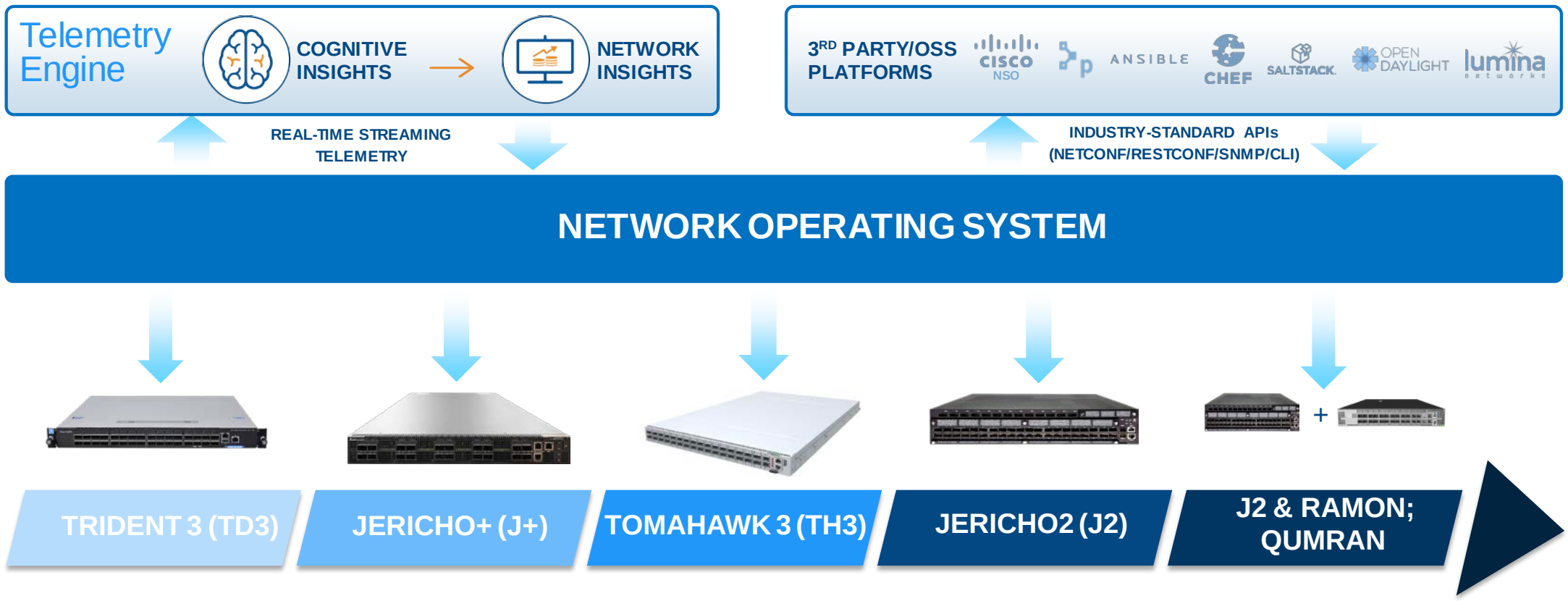
Open Integration: The Freedom to Thrive



Arrcus, Inc Confidential - Light Reading Webinar

ARRCUS NETWORK DIFFERENT

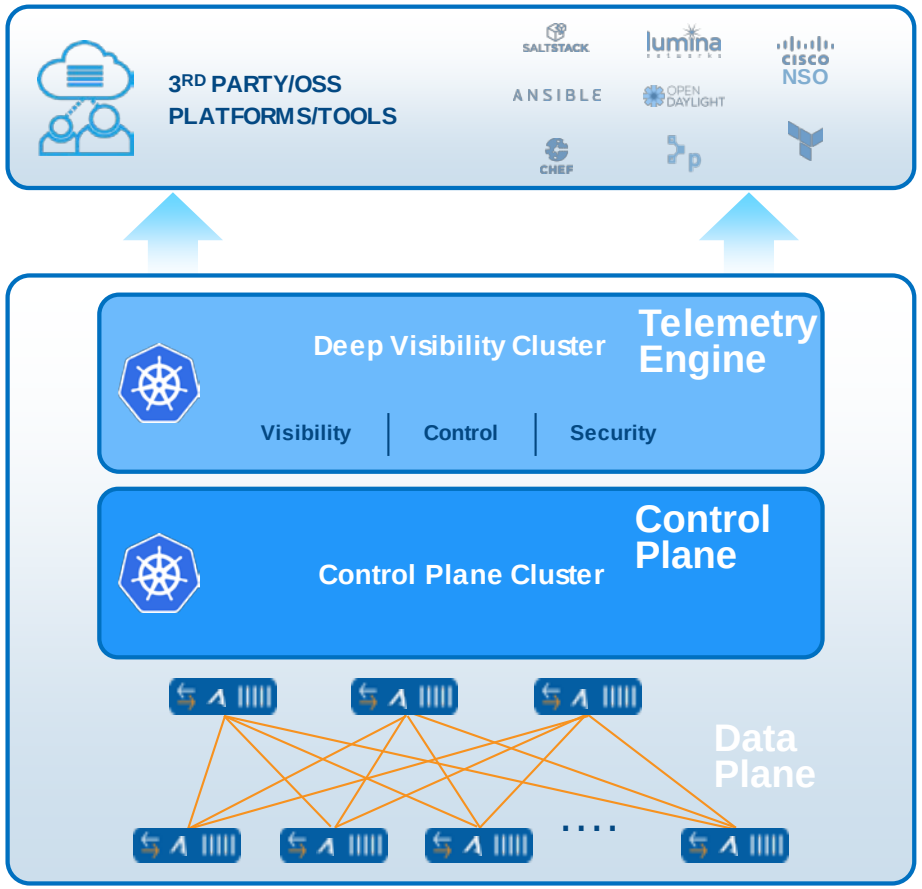
Powering the Broadest Range of State-of-the-Art Solutions



Arrcus, Inc Confidential - Light Reading Webinar

Virtualized Distributed Routing Solution

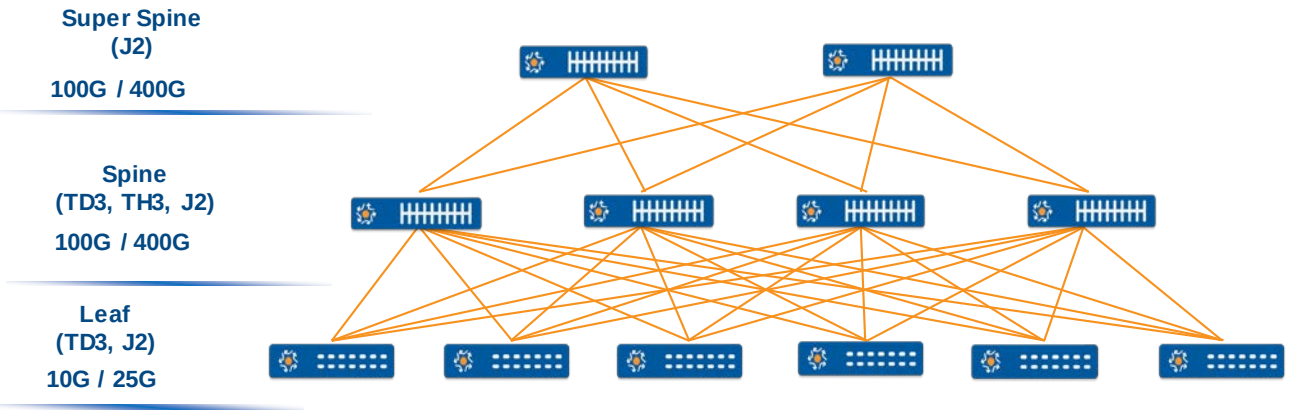
Virtualized, Distributed, Massively Scalable Router (VDR)



- Highly Available Control Plane at Internet Scale
- Single Point of Management & K8s-based Orchestration
- Massively scalable with 7680 100G ports
- Open, Standards-based Data Model: OpenConfig, CLI
- Deep Visibility & Analytics
- Lowest TCO: Pay-as-you-Grow

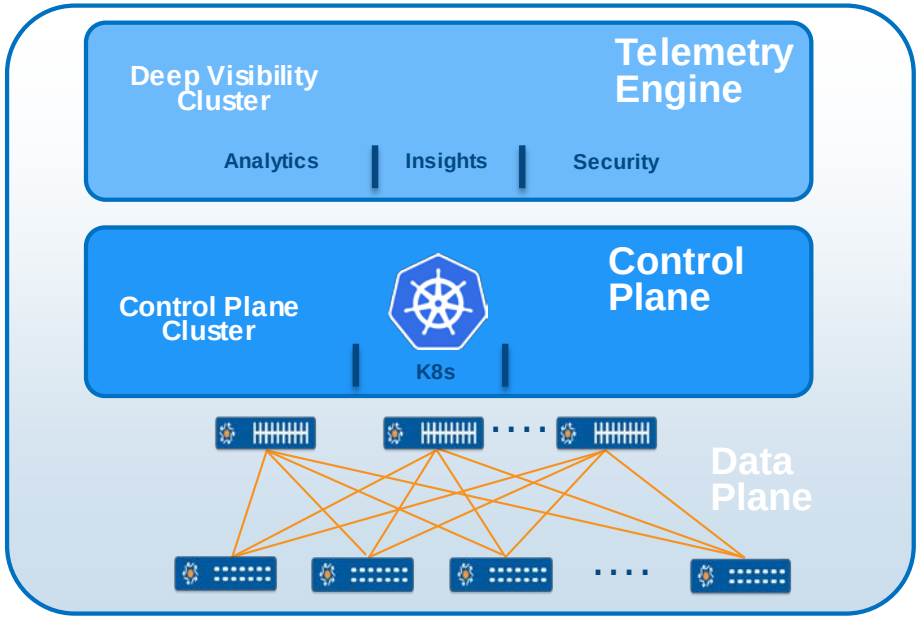
Arrcus, Inc Confidential - Light Reading Webinar

One Architecture: Limitless Possibilities



Distributed IP Clos

Distributed Control Plane, Distributed Data Plane



VDR

Virtualized Off-Box Control Plane, Distributed Data Plane

Single Architecture/Software

AnyService. AnyTime. AnyWhere

Telemetry Engine — COGNITIVE INSIGHTS → NETWORK INSIGHTS

REAL-TIME STREAMING TELEMETRY

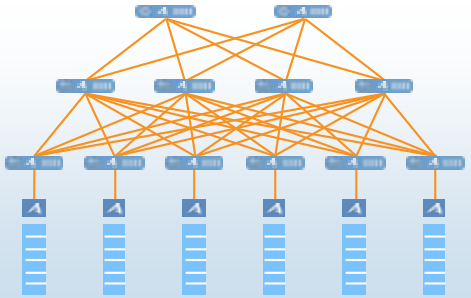
3RD PARTY/OSS PLATFORMS — cisco NSO, p, ANSIBLE, CHEF, SALTSTACK., OPEN DAYLIGHT, lumina

INDUSTRY-STANDARD APIs (NETCONF/RESTCONF/SNMP/CLI)

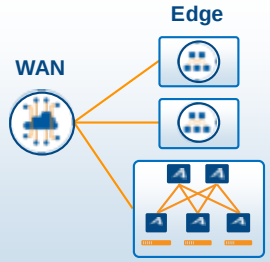
NETWORK OPERATING SYSTEM

BEST-IN-CLASS SOFTWARE: SIMPLE. SCALABLE. SEAMLESS

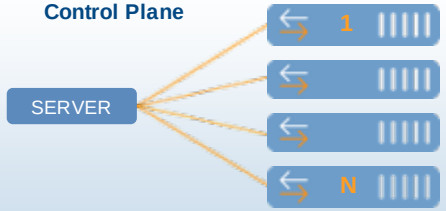
IP CLOS (ToR, LEAF, SPINE, SPLINE)



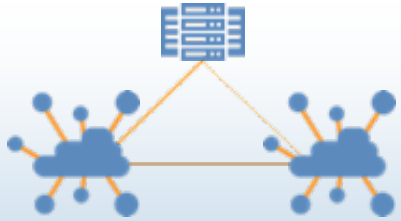
PEERING / EDGE



VIRTUALIZED DISTRIBUTED ROUTING



MULTI-CLOUD NETWORKING



ANY HARDWARE: ROUTERS. SWITCHES. SMART SERVERS; ANY DEPLOYMENT: ON-PREM. CONTAINER. CLOUD

FLEXIBLE CONSUMPTION MODEL. LOWEST TCO

Arrcus, Inc Confidential - Light Reading Webinar

ARRCUS NETWORK DIFFERENT

The Power of One



- ***One simple network***
- ***One scalable architecture***
- ***One seamless experience to connect billions***

ARRCUS
NETWORK DIFFERENT™

Thank You!

www.arrcus.com

